Queue Implementation Using Array

C Program Code and Execution Output

PROGRAM SOURCE CODE

```
#include<stdio.h>
#define MAX 50

int queue[MAX];
int rear = -1;
int front = -1;

void enqueue(int data)
{
    if(rear == MAX - 1)
        printf("Queue Overflow \n");
    else
    {
        if(front == -1)
        {
            front = 0;
        }
        rear = rear + 1;
        queue[rear] = data;
        printf("Element inserted is : %d\n", queue[rear]);
    }
}

void dequeue()
{
    if(front == -1 || front > rear)
    {
        printf("Queue Underflow \n");
        return;
    }
    else
    {
        printf("\nElement deleted is : %d\n", queue[front]);
        front = front + 1;
    }
}

void display()
{
    int i;
    if(front == -1) printf("Queue is empty \n");
    else
    {
        printf("Queue is : \n");
        for(i = front; i <= rear; i++)
            printf("%d ", queue[i]);
    }
}

int main()
{
    enqueue(10);
    enqueue(20);
```

PROGRAM OUTPUT

```
Element inserted is : 10
Element inserted is : 20
Element inserted is : 30
Element inserted is : 50
Element inserted is : 50
Element inserted is : 60
Queue is :
10 20 30 50 50 60
Element deleted is : 10

Element deleted is : 20
Queue is :
30 50 50 60
```